

MAHMOUD KHALED

Computer Science Student

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ABOUT ME

I'm a Computer Science student with a strong interest in software engineering and a passion for building practical, full-stack applications from the ground up. I enjoy working across the entire stack, from designing clean and responsive front-end interfaces to architecting reliable back-end systems and databases. I'm especially drawn to problem solving, system design, and writing efficient, well-structured code, and I'm always looking for opportunities to learn new technologies, contribute to open-source projects, and grow as a developer. I thrive in collaborative environments and enjoy turning ideas into functional, real-world products that genuinely solve problems for the people using them.

EDUCATION

MIU International University — Cairo, Egypt

Bachelor of Computer Science (2024 – Present)

EXPERIENCE

Software Development Intern — El-Zatuna 1 Month

- Gained hands-on experience contributing to real-world software development tasks within a professional team environment.

Intern — CIB (Commercial International Bank) 1 Month

- Completed a one-month internship exposure to enterprise workflows and technology practices within the banking sector.

TECHNICAL SKILLS

Frontend: TypeScript, JavaScript (ES6), HTML5, CSS3, React, Next.js, Responsive Design

Backend: Node.js, Express.js, REST API Design, JWT Authentication, Middleware

Databases: PostgreSQL, Drizzle ORM, MongoDB (Mongoose), Database Management Systems

Core: C++, Java, Data Structures & Algorithms

DevOps: Docker, Kubernetes, GitHub Actions CI/CD

Testing & Tools: Vitest, Supertest, Git & GitHub

PROJECTS

Cliniqo – Clinic Management Platform

GitHub: github.com/Nameless770/Cliniqo

Next.js 16, React 19, TypeScript, PostgreSQL 17, Drizzle ORM, Zod, Docker

- Built a HIPAA-oriented clinic management app covering appointments, patient records, clinical notes, prescriptions, billing, and a patient portal, with separate staff and patient sign-in (password with lockout, Google OAuth, optional TOTP two-factor).
- Designed PHI-safe data access: every read and write of a patient record is audited in the same transaction, clinical data is never hard-deleted, amendments preserve prior versions, and every server action re-checks permissions server-side.
- Implemented appointment scheduling with double-booking protection and a break-glass emergency access path that is logged and flagged for review.
- Built a symptom-triage assistant where a deterministic emergency check always runs first, with pluggable engines (local rules, local LLM via Ollama, or OpenAI) and post-generation filtering that discards any reply naming a medicine, dose, or diagnosis.
- Set up a four-layer verification suite: structural security-invariant checks, integration tests against a real PostgreSQL database, end-to-end journeys run with JavaScript disabled, and an automated permission-matrix check, wired into GitHub Actions CI.

Nova Marketplace – Multi-Vendor E-Commerce Platform

GitHub: github.com/Nameless770/Nova-Marketplace

React 19, Node.js/Express 5, MongoDB (Mongoose), JWT, Anthropic/OpenAI SDK

- Built a full-stack, three-role (customer/seller/admin) marketplace as a modular monolith — React SPA + Express API over a MongoDB replica set — with 90+ REST endpoints across 18 route modules.
- Engineered concurrency-safe money and inventory logic: integer minor-unit pricing, prices re-verified at capture, and atomic conditional MongoDB updates that make overselling or over-refunding structurally impossible.
- Enforced server-side multi-tenant security and hardened JWT auth (pinned algorithm, issuer/audience checks, short-lived tokens, brute-force lockout) plus an append-only audit log on privileged mutations.
- Architected an AI shopping assistant and admin analytics tool with a "model selects, database renders" grounding contract, using schema-constrained outputs and post-generation fact re-verification to eliminate hallucinated prices or stock data.
- Delivered production-readiness tooling: 224 automated tests (Vitest/Supertest) including cross-tenant attack tests, structured logging and Prometheus metrics, and a GitHub Actions CI/CD pipeline with Docker image publishing.

BuzzMind – Full-Stack Web Platform

GitHub: github.com/Nameless770/BuzzMind

Built BuzzMind — a full-stack web platform for instructors and students to create and run live quizzes, manage assignments, and track results with real-time chat and leaderboards.

- Designed a full-stack architecture connecting a dynamic front-end to a Node.js/Express back-end with a MongoDB database
- Implemented live quiz sessions with real-time updates so instructors can run quizzes interactively in class
- Built real-time chat functionality to support communication between instructors and students during sessions
- Developed a leaderboard system to track and display student rankings and quiz performance
- Created assignment management features allowing instructors to post, track, and review student submissions

VOLUNTEERING

Co-Head, Software Development — MSP Tech Club, MIU

- Contributed to software development initiatives as part of the MSP Tech Club at MIU.